Leo Conforti

Software Developer I

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SUMMARY

Software Developer holding a Bachelor of Science in Computer Science with experience in fullstack development, astrophysics image processing, and distributed systems.

WORK EXPERIENCE

PREV SOFTWARE DEVELOPER CO-OP STUDENT | FACTORYTWIN **May 2023 - January 2024**

As a co-op student, I took a much lighter course load in the evenings and commuted to work full time 8am-4pm in person in Hopkins. I was tasked with building both our constrained and unconstrained (infinite capacity) schedulers, whose outputs are used to derive the predictive analytics of FactoryTwin.

PREV SOFTWARE DEVELOPER INTERN | FACTORYTWIN **May 2024 - August 2024**

As an intern, I built multiple new charts and pages in FactoryTwin, contributing heavily to the brand new Continuous Improvement and Validator modules of FactoryTwin. I decided not to take more time off of school to do another nine month co-op in order to guarantee that I could still graduate in four years.

SOFTWARE DEVELOPER I | FACTORYTWIN **August, 2025 - Present**

- Designed software architecture and database schema
- Formalized factory physics into a robust and generalized data model representing factory operations
- Delivered to onpremise machines and maintained testing environments
- Read in customer data from varying sources and built ETLs to automated that process
- Optimized our predictive analytics scheduler algorithms and reduced runtimes from hours to under 10 minutes

UNDERGRADUATE RESEARCHER | UNIVERSITY OF MINNESOTA COLLEGE OF SCIENCE AND ENGINEERING **January, 2025 - May, 2025**

https://twin-cities.umn.edu/news-events/u-m-leading-1-million-grant-build-superfast-turbo-telescopes

Our group, TURBO, focused on observing neutron star collisions in collaboration with NMU.

- Built and administered six high performance gpu computers
- Implemented astrophysics algorithms in the image processing pipeline
- Upgraded and performed migrations on existing relational databases
- Reduced query execution time on existing databases by 50%
- Built and owned the web analytics dashboard which displays metrics and results of the pipeline runs.
- Trained and evaluated convolutional neural networks (CNNs) to automate transient image classification and improve detection accuracy.

COMPUTER SCIENCE TA | UNIVERSITY OF MINNESOTA COLLEGE OF SCIENCE AND ENGINEERING **September, 2024 - December, 2024**

Undergraduate TA for CSCI 4271W Development of Secure Software in the computer science department of the University of Minnesota

- Graded assignments, labs, reports, and exams providing help and feedback during my office hours
- Assisted and ran lab sections
- Reviewed curriculum

SKILLS

IT Infrastructure Management • IT Infrastructure Design • FullStack Development • Image Processing • High Performance Computing (HPC) • Data Structures • Algorithms • Algorithm Optimization • Network Security • Cybersecurity • Information Security

SOFT SKILLS

Communication and collaboration with teammates • Adaptability to changing project requirements • Time management and task prioritization

EDUCATION

Bachelor of Science | Computer Science

University of Minnesota

September, 2021 - May, 2025